

Prevention of Phone Scams Through the Identification of Voice Prints

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Abstract - The development of generative AI has contributed to a new era of creativity and an era of impersonation and identity threats. The rise of voice cloning and real-time voice conversion has led to phishing crimes and ethical vulnerabilities. Cybercriminals have exploited this technology to impersonate individuals, resulting in fraud, identity theft, and phone scams, often referred to as voice phishing. This thesis proposes a machine learning model to prevent phone scams and phishing crimes by using voice recognition technology to analyze voice data and detect manipulated voices. By analyzing different audio features, the proposed model, utilizing the Random Forest algorithm, achieved an accuracy of 95.93%. This paper confirms that the Random Forest algorithm yields high detection rates and strong performance with real-time audio conversion capabilities—crucial given that voice phishing occurs in real time. This study presents a solution to prevent cyber crimes caused by voice deep fakes through the use of machine learning.

Keywords: *Fake Voice Detection, Voice Recognition, Manipulated Voices, Real-Time Detection, Machine Learning, Random Forest*

I. INTRODUCTION

The emergence of AI-generated media, particularly Deepfake technology, has introduced new challenges in cybersecurity. From its original purpose of entertainment, Deepfake technology has evolved to manipulate both visual and auditory content with high precision. In voice synthesis, Deepfake audios, also known as "synthetic voices," have emerged as a significant concern. Cybercriminals have exploited this technology to impersonate individuals, leading to fraud, identity theft, and phone scams, often referred to as voice phishing crimes [1].

Voice phishing involves the use of manipulated or AI-generated voices to deceive victims and obtain sensitive information, such as financial credentials. With advances in AI, these synthetic voices can now mimic not only the speech patterns but also the intonation, cadence, and emotional tone of the target voice, making detection by human listeners increasingly difficult. As such, addressing Deepfake audio poses a growing challenge in cybersecurity.

Current Research and Challenges

Several methods have been proposed to address the detection of Deepfake audio. Early research focused on analyzing the physical characteristics of audio signals, such as pitch, rhythm, and speech modulation, but these methods had limited success against sophisticated AI-generated voices. More recently, deep learning models,

including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have been developed to detect subtle anomalies in voice data that are imperceptible to the human ear.

For instance, a study by Zhuang (2023) proposed a CNN-based model for voice-changing detection, achieving an average detection accuracy of 67.81% across different tonal ranges in English and Chinese languages. Additionally, the work of Bird and Lotfi (2023) suggests real-time detection methods that leverage advanced machine learning architectures, such as the Retrieval-based Voice Conversion (RVC) model, to identify Deepfake voices.

Despite the advancements in Deepfake audio detection, current approaches have several shortcomings. First, the lack of large, diverse, and high-quality datasets of both real and Deepfake audio limits the generalizability of detection models. Most available datasets are either artificially generated or too small to capture the wide range of variations seen in real-world scenarios. This limitation has led to models that are prone to overfitting and cannot perform well when tested on new, unseen data.

Second, while CNN and RNN models have demonstrated some success, they tend to require substantial computational resources, making real-time detection difficult in resource-constrained environments such as mobile devices. Furthermore, while XAI provides useful insights into model decisions, its integration into existing audio detection frameworks remains a challenge due to the complexity of Deepfake audio manipulation [2].

Another significant limitation is that many models primarily focus on accuracy metrics such as precision and recall but overlook practical implementation issues such as processing speed and resource efficiency. This gap between theoretical performance and practical application has hindered the deployment of these models in real-time voice phishing prevention systems.

This thesis proposes a more efficient solution for detecting Deepfake audio in real-time by leveraging the Random Forest algorithm. Unlike deep learning models that require high computational power, the Random Forest algorithm offers comparably high detection accuracy while maintaining efficiency in both training and real-time applications.

Model Proposal

This paper aims to produce a model of voice phishing that can detect voice manipulation and deep fakes. The paper focuses on testing the feasibility of machine learning as a solution to voice phishing, finding a model with a

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relatively low number of calculations with high accuracy of voice phishing classification.

II. METHODS

Dataset

The data used in this study was obtained from Kaggle's DEEP-VOICE: DeepFake Voice Recognition dataset, which contains a total of 64 audio samples—8 real voice samples and 56 fake voice samples [3]. The features extracted from the audio data include Chromagram, Root Mean Square (RMS), Spectral Centroid, Spectral Bandwidth, Rolloff, Zero Crossing Rate, and Mel-Frequency Cepstral Coefficients (MFCC 1 through 20). This dataset consists of a total of 62 minutes and 22 seconds of speeches, from eight public figures, with their voices converted into one another using Retrieval-based Voice Conversion to simulate real-time voice manipulation.

Feature Extraction

From the audio dataset, the audio was broken into 1-second blocks and the audio features used included Chroma Short-Time Fourier Transform (STFT), Root Mean Square (RMS) Energy, Spectral Centroid, Spectral Bandwidth, Spectral Rolloff, Zero Crossing Rate (ZCR), and Mel Frequency Cepstral Coefficients (MFCC).

STFT represents the energy distribution across the 12 pitch classes, used to analyze the harmonic content of audio. RMS Energy measures the signal's power and its perceived loudness. Spectral Centroid identifies the center of the spectrum, associated with the timbre of the sound, while Spectral Bandwidth quantifies the spread of frequencies around this centroid. Spectral Rolloff is the frequency below which a certain percentage of the total spectral energy lies, giving insight into the distribution of energy across frequencies. ZCR counts how frequently the signal crosses the zero axis, reflecting the signal's noisiness. Finally, MFCC captures the short-term power spectrum on a Mel scale, mimicking human auditory perception, and is widely used in speech recognition tasks.

Model

As a result of reviewing similar studies, it has been determined that DeepFake voice data generated by artificial intelligence can be classified with the highest performance and accuracy using machine learning models. The XGBoost algorithm ranks first in terms of performance, followed by Random Forest in second place, as illustrated in Fig. 1 [3].

Although XGBoost yielded the most accurate and reliable results, it is computationally intensive, as it optimizes both accuracy and speed by significantly increasing the computational load. Given the constraints of the study, which assumed the absence of a GPU capable of supporting such high computational demands, XGBoost was excluded from further analysis. Instead, the Random Forest algorithm was selected, as it provides comparable results while requiring substantially lower computational resources.

Model	Mean Value over 10-fold Cross Validation						Inference Time (ms)
	Acc.	Prec.	Rec.	F1	MCC	ROCAUC	
XGBoost (310)	0.993	0.995	0.991	0.993	0.986	0.993	0.004
Random Forest (310)	0.989	0.995	0.983	0.989	0.978	0.989	0.057
Quadratic Discriminant Analysis	0.948	0.969	0.924	0.946	0.896	0.948	0.002
Linear Discriminant Analysis	0.889	0.886	0.893	0.889	0.778	0.889	0.001
Ridge	0.883	0.884	0.882	0.883	0.767	0.883	0.001
Naive Bayes (Gaussian)	0.830	0.864	0.784	0.822	0.664	0.830	0.001
KNN (1)	0.815	0.808	0.827	0.817	0.630	0.815	0.143
SVM	0.723	0.815	0.576	0.675	0.465	0.723	0.605
Naive Bayes (Bernoulli)	0.692	0.742	0.587	0.655	0.391	0.691	0.001
Stochastic Gradient Descent	0.668	0.732	0.760	0.681	0.407	0.673	0.001
Gaussian Process	0.614	0.997	0.229	0.372	0.358	0.614	0.561

Figure 1. Comparison of Machine Learning Models in Detecting DeepFake Voice Data [3]

The Random Forest algorithm is a machine-learning algorithm that combines the output of multiple decision trees to reach a single result. The random forest algorithm is an extension of the bagging method as it utilizes both bagging and feature randomness, which generates a random subset of features ensuring low correlation among decision trees.

The decision trees are frequently utilized in machine learning, but they often encounter issues such as bias and overfitting. However, the Random Forest algorithm mitigates such issues as when multiple decision trees combine to form an ensemble, the algorithm yields notably improved predictions, especially when the individual trees within it maintain a lack of correlation with one another. A diagram of how the Random Forest algorithm works can be found below in Fig. 2.

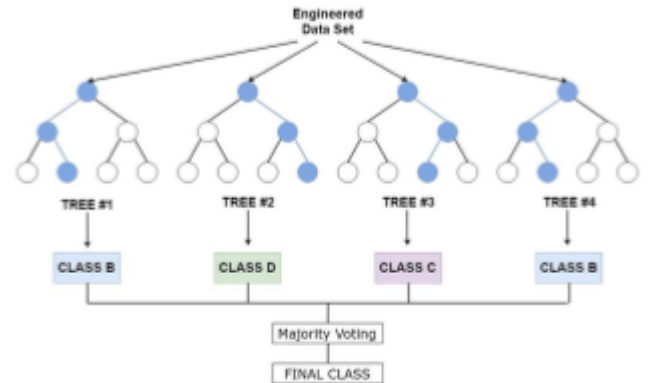


Figure 2. Visualization of Random Forest Classification

In previous studies, 10-fold cross-validation was used, where the dataset was divided into ten subsets, and the model was trained on 9/10th and tested on 1/10th of the data, iteratively[3]. While effective, this approach requires training and testing the model ten times, demanding significant time and computational resources. In contrast, this study adopts a more resource-efficient method by using a 70:30 training-to-test data split, a common practice that reduces overfitting by isolating a separate test set to assess the model's performance on unseen data. This approach helps detect overfitting and ensures generalization when the model encounters new input.

III. RESULTS

The experiments were conducted on a MacBook Pro (13-inch, M1, 2020) and on Google Colab, using Python 3.11. The Librosa library was employed for visualizing voice data files and extracting relevant features, while Pandas was utilized to store the extracted data in data frames. The primary model for training the data was implemented using the RandomForest library. For validation, the dataset was split into training and testing sets in a 7:3 ratio, with dynamic randomization to ensure proper allocation. To evaluate model performance, the sklearn.metrics library was used for scoring, and the results were visualized using matplotlib.

Results of Model

The study achieved 95.93% accuracy with all the voices that were used. The difference between this result and other previous studies showed that the quantified results depended on the method of validation[3].

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

To evaluate the model, accuracy, precision, and F1 score were used as evaluation metrics. Accuracy measures the proportion of correctly predicted instances among all predictions but can be misleading if the data is imbalanced, calculated by equation (1). Precision measures how often the model correctly predicts the positive class, calculated by equation (2). F1-score is the harmonic mean of precision and recall, balancing them when there's an uneven class distribution, calculated by equation (3). By the results of these evaluation metrics displayed by Fig. 3, the high accuracy, precision, and F1-Score indicate an effective model in detecting Deep fake voices.

TABLE I. RESULTS OF MODEL ON EVALUATION METRICS

Evaluation Metrics	Average	Median	Standard Deviation
Accuracy	95.93%	96.21%	2.02%
Precision	0.96	0.96	0.02
F1-Score	0.96	0.96	0.02

Figure 3. Table of Results for Accuracy, Precision, and F1 score.

IV. DISCUSSION

By creating a binary classification model through the Random Forest algorithm, this study presents a practical solution for preventing voice phishing, addressing prior research gaps in detection accuracy and computational efficiency. With an impressive accuracy of 95.93%, precision of 0.96, and F1-score of 0.96, this model offers a commercial, scalable tool for voice phishing detection, particularly in resource-constrained environments. While existing models like CNNs and RNNs require substantial computational power, the Random Forest algorithm strikes

a balance between accuracy and low computational demands, making it ideal for real-time applications.

However, it is essential to acknowledge the data limitations in this field. Since voice data from phishing crimes are rarely available publicly, further research is required to improve the model's generalizability. Expanding access to more extensive datasets could improve training and increase the model's applicability across diverse scenarios.

A potential extension of this project involves the development of a mobile app that provides real-time voice phishing detection. This app would warn users instantly of potential phishing threats using the efficient Random Forest-based model, offering a high performance tool for protecting individuals from voice-based scams. Such an application has the potential to reduce the occurrence of phone scams, increasing security in everyday voice interactions.

REFERENCES

- [1] Cho, H.-D. (2012). Voice phishing occurrence and counterplan. Journal of the Korea Contents Association, 12(7), 176-182.
- [2] Zhuang, C. (n.d.). Voice-changing detection with convolutional neural network. Retrieved December 21, 2023, from https://listings.lib.msu.edu/repo/Voice-Changing_Detection_with_Convolutional_Neural_Network.pdf
- [3] Bird, J. J., & Lotfi, A. (2023). Deepfake audio detection and justification with Explainable Artificial Intelligence (XAI). Retrieved from https://www.researchgate.net/publication/374798687_Deepfake_audio_detection_and_justification_with_Explainable_Artificial_Intelligence_XAI
- [4] Xu, Y., Fu, Z., Yu, X., Li, J., & Wang, X. (2022). A survey of deepfake audio detection methods: From traditional machine learning to deep learning approaches. Applied Sciences, 12(8), 3926. <https://doi.org/10.3390/app12083926>
- [5] Prashanth, G., Varun, S., & Raman, R. (2023). Real-time detection of AI-generated speech for DeepFake voice conversion. arXiv preprint. <https://arxiv.org/pdf/2308.14970>
- [6] Kim, J. W., Salamon, J., Li, P., & Bello, J. P. (2018). CREPE: A convolutional representation for pitch estimation. In Speech and Signal Processing (ICASSP) (pp. 161-165). IEEE.